FarmLens

Your land, your climate, your best crops, with the reasons.

NSW land screening that puts crop possibilities, evidence and local questions in one view.

CODEX
BUILDER DAY

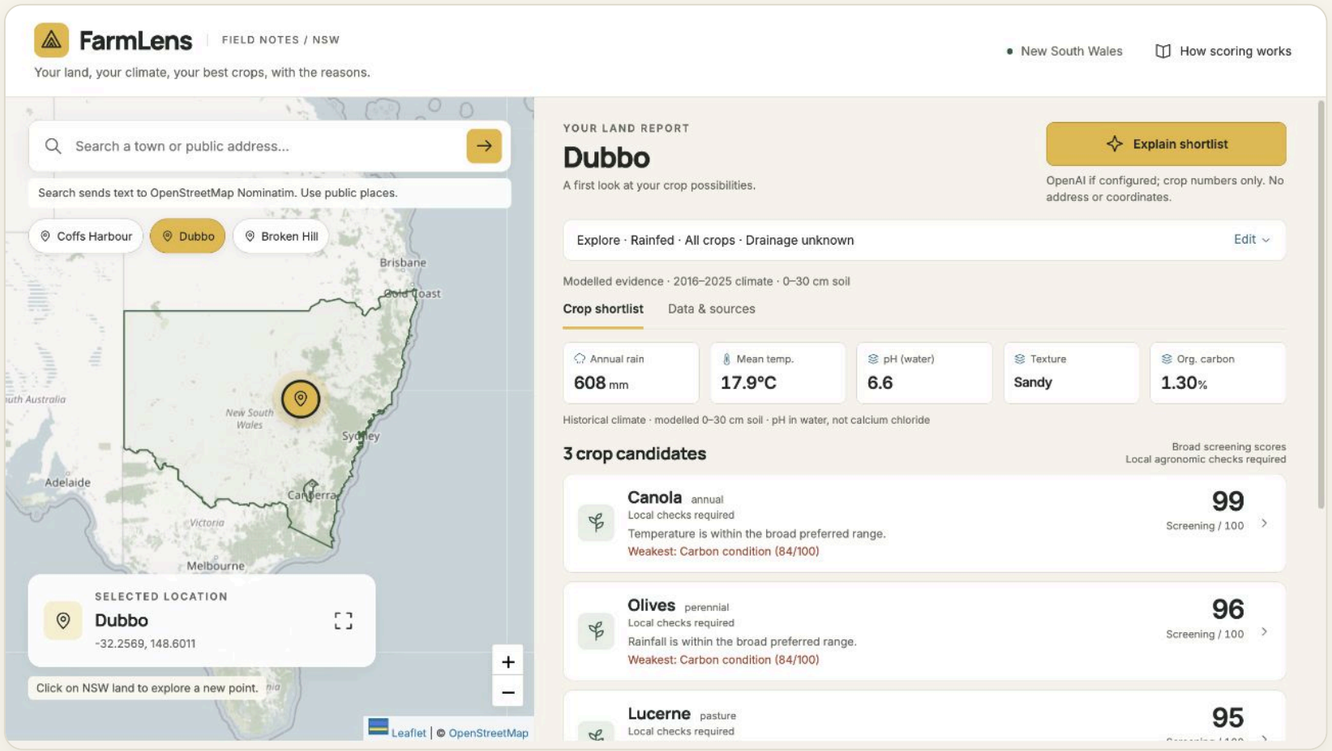
PROJECT
SHOWCASE



01 Choose a NSW point · **02** Explore or plan a season · **03** Inspect the reasons · **04** Compare a water scenario

FROM SCATTERED DATA TO A READABLE SHORTLIST

Recorded custom interface · Dubbo



15 crop profiles · factor bars · source records · plain-English explanations. A score is an index, not a success probability.

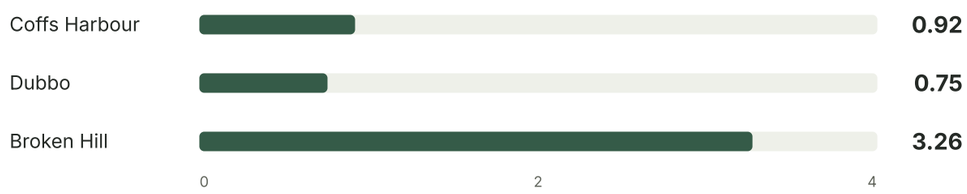
DEMO INSIGHT / SEPARATE WATER PILOT

A crop score is only the start of the water question.

Compare wheat, barley and canola across ten complete historical seasons. Change the assumptions; inspect the water gap. Crop rankings stay unchanged.

Same wheat scenario. Different locations.

Median estimated gross irrigation · ML/ha



May 1 sowing, 2016-2025 seasons. Illustrative inputs: 100 mm soil-water buffer; 50 mm at sowing; 80% rain infiltration and irrigation efficiency; 150-day crop duration.

Historical scenarios, not irrigation prescriptions. Wheat and barley match under identical default inputs.

01 / EVIDENCE & METHOD

Make the reasoning inspectable.

ERA5 climate + modelled CSIRO/TERN SLGA soil + approximate FAO EcoCrop profiles. Seven weighted factors and source-linked NSW checks expose constraints.

Water pilot: daily rain and FAO-56 ET0 × crop coefficients, with an editable storage balance.

02 / ENGINEERING

Designed to show uncertainty.

Python engine + custom web UI. Missing soil pH withholds the shortlist; absent factors stay unassessed. Cached evidence retains source status.

230 Python + 11 JavaScript tests in the recorded pre-showcase run. Software checks are not agronomic validation.

03 / LIMITS & NEXT STEPS

From prototype to local validation.

Modelled inputs and illustrative water defaults are not farm measurements. No yield or profit forecasts. Scores and water estimates are uncalibrated.

Next: agronomist review, measured soil-water comparisons and locally calibrated crop timing.

Sources: Open-Meteo ERA5 · CSIRO/TERN SLGA (CC BY 4.0) · FAO EcoCrop · FAO-56 · linked NSW guidance.
Map © OpenStreetMap contributors (ODbL); boundary © ABS (CC BY 4.0). Example photo: "Canola field temora nsw.jpg", John O'Neill (edited by Fir0002), CC BY-SA 3.0; cropped.
Evidence snapshot: 20 October 2026. Full method, credits and reproducibility notes accompany this poster in sources-and-notes.md.

Indicative only.
Check with a local agronomist.